

TØ 9 – Mikroøkonomi 2 – 2024

Første opgave er fra PS8, men blev gennemgået den 10. april sammen med PS9, fordi vi ikke havde tid til TØ 8

Insurance

A risk averse individual (agent A) with initial wealth w has a new car valued L subject to a risk of being stolen. In order to protect himself in the case of the car being stolen, the individual can buy insurance from a risk neutral insurance company (principal P). The company and the individual share the same common prior of the risk of the car being stolen. In particular the car is stolen with probability k , which we here assume constant and independent of any actions of the agent.

The individual chooses coverage q such that he either insures fully ($q = L$) or only partly ($q < L$). The insurance company demands a price (or a premium) p for providing insurance coverage q . The price is paid regardless of whether the car is stolen or not.

The car is valued as an asset to the individual and thus affects his utility through the impact on his total wealth, i.e. in the absence of any insurance the agent gets utility $u(w)$ when the car is stolen and $u(w + L)$ when the car is not stolen.

1. Determine A's *reservation utility* (outside option). I.e. the utility A gets from not participating in the insurance market.
2. What could P include as verifiable terms in the insurance contract? (what does the contract depend on)
3. Assume P decides on p and q . State the optimization problem and derive FOC.
4. Solve for the q and p which the principal chooses in optimum. Do P earn positive profits? Is the outcome efficient?

1

Reservation utility is

$$\bar{u} = k \cdot u(w) + (1 - k) \cdot u(w + L)$$

That is, the probability k of the car being stolen times his utility. In other words: he gains L in utility from the car not being stolen

2

The price (or premium) offered by the insurance company is verifiable. So is the coverage chosen by the agent. Factors such as initial wealth, car's value, and the probability of it being stolen are of course not directly observable to the insurance company.

3

Maximization problem of the principal is

$$\begin{aligned} \max(1 - k) \cdot p + k(p - q) &= p - kq \\ \text{s.t. } k \cdot u(w - p + q) + (1 - k)u(w + L - p) &\geq \bar{u} \end{aligned}$$

That is, the probability of the car not being stolen, and the car being stolen must be chosen such that it is equal to the price paid by the costumer minus the probability of it being stolen times the coverage the insurance company must pay if it happens.

Subject to the PC, i.e., the agent must get a higher utility be accepting the insurance offer than if he did not ($\bar{u}$ is defined in 1).

Lagrange:

$$\mathcal{L} = p - kq + \lambda((1 - k)u(w + L - p) + k \cdot u(w - p + q) - \bar{u})$$

FOCs are

$$\frac{\partial \mathcal{L}}{\partial p} = 1 - \lambda((1 - k)u'(w + L - p) + ku'(w - p + q)) = 0$$

$$\frac{\partial \mathcal{L}}{\partial q} = -k + \lambda(ku'(w - p - q)) = 0$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = (1 - k)u(w + L - p) + k \cdot u(w - p + q) = 0$$

4

FOCs can be rewritten into

The second one:

$$-k + \lambda(ku'(w - p - q)) = 0 \Leftrightarrow \lambda = \frac{1}{u'(w - p - q)}$$

So, we can substitute the value for λ into the first one

$$\begin{aligned} 1 - \lambda((1 - k)u'(w + L - p) + ku'(w - p + q)) \\ = 1 - \frac{1}{u'(w - p - q)}((1 - k)u'(w + L - p) + ku'(w - p + q)) = 0 \Leftrightarrow \end{aligned}$$

$$1 = \frac{1}{u'(w - p - q)} ((1 - k)u'(w + L - p) + ku'(w - p + q))$$

And now, since (in the first equation), we can insert the value of lambda

$$\begin{aligned} 1 - \lambda((1 - k)u'(w + L - p) + ku'(w - p + q)) \\ = 1 - \frac{1}{u'(w - p - q)} ((1 - k)u'(w + L - p) + ku'(w - p + q)) = 0 \\ \Leftrightarrow \frac{1}{u'(w - p - q)} ((1 - k)u'(w + L - p) + ku'(w - p + q)) = 1 \\ \Leftrightarrow u'(w - p - q) = (1 - k)u'(w + L - p) + ku'(w - p + q) \end{aligned}$$

But the above equation implies that

$$u'(w - p - q) = u'(w + L - p)$$

So therefore, $q = L$

This can be used in the PC (binding since in optimum)

$$\begin{aligned} (1 - k)u(w + L - p) + k \cdot u(w - p + q) &= (1 - k) \cdot u(w + L - p) + k \cdot u(w - p + L) \\ &= u(w + L - p) \\ &= \bar{u} \\ \Leftrightarrow w + L - p &= u^{-1}(\bar{u}) \\ \Leftrightarrow -w - L + p &= -u^{-1}(\bar{u}) \\ p &= w + L - u^{-1}(\bar{u}) \end{aligned}$$

Profits are then

$$p - kq = w + L - u^{-1}(\bar{u}) - k \cdot L = w + (1 - k)L - u^{-1}(\bar{u}) > 0$$

Since $p = w + L - u^{-1} \cdot \bar{u} > 0$ by definition, and therefore $p - kq = w + (1 - k)L - u^{-1}(\bar{u}) > 0$ as we do not have perfect competition (just yet)

It is efficient since the agent receives full coverage and the principal's profits are positive

5. Assume P competes in a perfectly competitive market with other insurance companies. What price p would P charge?
6. What coverage q would A prefer with the price being set as in 5)? Is the outcome efficient?
7. What contract does P offer if A is risk neutral ($u(w) = w$)?
8. Assume that both P and A are risk averse (P 's utility function is $W(\pi)$). Would P offer full coverage $q = L$?

5

In a perfect market, the profit will be zero (just basic micro 1) and hence,

$$\max(1 - k) \cdot p + k(p - q) = p - kq = 0 \Rightarrow p = kq$$

6

Since the agent is risk averse, he will choose full coverage, so the maximisation of the agent is

$$\max ku(w + q - kq) + (1 - k)u(w + L - kq)$$

Lagrange (not really necessary, just find the derivative wrt q)

$$\mathcal{L} = ku(w + q - kq) + (1 - k)u(w + L - kq)$$

FOC is

$$\frac{\partial \mathcal{L}}{\partial q} = -(1 - k)u'(w + L - kq)k + (1 - k)ku'(w - kq + q) = 0$$

Which can be simplified:

$$\begin{aligned} & -(1 - k)u'(w + L - kq)k + (1 - k)ku'(w - kq + q) \\ & = (-1 + k)u'(w + L - kq)k + (1 - k)ku'(w - kq + q) \\ & = u'(w + L - kq)k + ku'(w - kq + q) \\ & = u'(w + L - kq) + ku'(w - kq + q) = 0 \\ & \Leftrightarrow u'(w + L - kq) = u'(w - kq + q) \Rightarrow L = q \end{aligned}$$

That is, no matter the probabilities, he will always choose full coverage. If it is stolen, he pays price kq but receives q in premium. If it is not stolen, he gains the L (his valuation) but still pays price kq and, needless to say, does not receive the premium.

It is efficient since q must be chosen such that $q \leq L$ which is satisfied

7

If the agent is risk-neutral, $u(w) = w$, we can simplify the reservation utility into

$$\bar{u} = k \cdot u(w) + (1 - k) \cdot u(w + L) = kw + (1 - k)(w + L) = w + (1 - k)L$$

Utility from accepting the offer is

$$\begin{aligned} (1 - k) \cdot u(w + L - p) + k \cdot u(w - p + q) &= (1 - k)(w + L - p) + k \cdot (w - p + q) \\ &= w - p + (1 - k) \cdot L + k \cdot q \end{aligned}$$

And we insert this into the PC:

$$w - p + (1 - k) \cdot L + k \cdot q \geq \bar{u} = w + (1 - k)L \Leftrightarrow kq \geq p$$

What does this mean?

When $kq > p$, negative profits

When $kq = p$, the firm earns zero profit regardless of the level of coverage. Thus, both the firm and the agent are indifferent between each level of coverage.

8

π and p are equivalent. Very confusing I know. I will suggest that the description should be changed

$$\begin{aligned} \max & (1 - k)W(p) + kW(p - q) \\ \text{s.t.} & (1 - k)u(w + L - p) + ku(w - p + q) \geq \bar{u} \end{aligned}$$

Lagrangian

$$\mathcal{L} = (1 - k)W(p) + kW(p - q) + \lambda[(1 - k)u(w + L - p) + ku(w - p + q) - \bar{u}]$$

FOCs

$$\frac{\partial \mathcal{L}}{\partial p} = (1 - k)W'(p) + kW'(p - q) - \lambda[(1 - k)u'(w + L - p) + ku'(w - p + q)] = 0$$

$$\frac{\partial \mathcal{L}}{\partial q} = -kW'(p - q) + \lambda[ku'(w - p + q)] = 0$$

Rewrite the FOCs

$$(1 - k)W'(p) + kW'(p - q) = \lambda[(1 - k)u'(w + L - p) + ku'(w - p + q)]$$

$$kW'(p - q) = \lambda[ku'(w - p + q)]$$

Combine the two:

$$\frac{(1-k)W'(p) + kW'(p-q)}{kW'(p-q)} = \frac{\lambda[(1-k)u'(w+L-p) + ku'(w-p+q)]}{\lambda[ku'(w-p+q)]}$$

Simplify the LHS

$$\frac{(1-k)W'(p) + kW'(p-q)}{kW'(p-q)} = \frac{(1-k)W'(p)}{kW'(p-q)} + \frac{kW'(p-q)}{kW'(p-q)} = \frac{(1-k)W'(p)}{kW'(p-q)} + 1$$

And RHS

$$\begin{aligned} \frac{\lambda[(1-k)u'(w+L-p) + ku'(w-p+q)]}{\lambda[ku'(w-p+q)]} &= \frac{(1-k)u'(w+L-p)}{ku'(w-p+q)} + \frac{ku'(w-p+q)}{ku'(w-p+q)} \\ &= \frac{(1-k)u'(w+L-p)}{ku'(w-p+q)} + 1 \end{aligned}$$

Putting these together yields

$$\frac{(1-k)W'(p)}{kW'(p-q)} + 1 = \frac{(1-k)u'(w+L-p)}{ku'(w-p+q)} + 1$$

$$\Leftrightarrow \frac{(1-k)W'(p)}{kW'(p-q)} = \frac{(1-k)u'(w+L-p)}{ku'(w-p+q)}$$

$$\Leftrightarrow \frac{W'(p)}{W'(p-q)} = \frac{u'(w+L-p)}{u'(w-p+q)}$$

The last line: multiply by $\frac{k}{1-k}$ on both sides of the equality

Would P then offer full coverage, i.e., $q = L$?

$$\begin{aligned} \frac{W'(p)}{W'(p-q)} = \frac{u'(w+L-p)}{u'(w-p+q)} &\Rightarrow \frac{W'(p)}{W'(p-L)} = \frac{u'(w+L-p)}{u'(w-p+L)} = 1 \Leftrightarrow W'(p) = W'(p-L) \\ &\Leftrightarrow L = 0 \end{aligned}$$

But $L > 0$, so P will not offer $q = L$

9. Assume $\frac{k}{1-k} = 1$, $U(w) = \ln(w)$, $W(\pi) = \ln(\pi)$, $W=10$ and $L=5$. What coverage would P offer?

Use the modified FOC to obtain:

$$\frac{W'(p)}{W'(p-q)} = \frac{u'(w+L-p)}{u'(w-p+q)} \Leftrightarrow \frac{p}{p-q} = \frac{w+L-p}{w-p+q} \Leftrightarrow p = q \left(\frac{w+L}{L} \right) = q \left(\frac{10+5}{5} \right) = 3q$$

We use the binding PC (because PC binds in optimum) to solve for p

LHS of the PC is

$$(1-k)u(w+L-p) + ku(w-p+q) = (1-k)\ln(w+L-p) + k\ln(w-p+q)$$

And RHS, the reservation utility:

$$\bar{u} = k \cdot u(w) + (1-k) \cdot u(w+L) = k \cdot \ln(w) + (1-k) \cdot \ln(w+L)$$

Putting the two together

$$\begin{aligned} (1-k)\ln(w+L-p) + k\ln(w-p+q) &= k\ln(w) + (1-k)\ln(w+L) \\ \Leftrightarrow (1-k)\ln(w+L-3q) + k\ln(w-3q+q) &= k\ln(w) + (1-k)\ln(w+L) \\ \Leftrightarrow (1-k)\ln(15-3q) + k\ln(10-2q) &= k\ln(10) + (1-k)\ln(15) \\ \Leftrightarrow \ln(15-3q) + \ln(10-2q) &= \ln(10) + \ln(15) \Leftrightarrow q \in \{0, 10\} \end{aligned}$$

Last step: because $\frac{k}{1-k} = 1 \Rightarrow k = 1-k$

So, $q = 0$ or $q = 10$

Notice that the second solution yields $\ln(w+L-p) = \ln(w+L-3q) = \ln(10+5-3 \cdot 10) = \ln(-15)$ which is undefined. Hence, $q = 0$ is the only valid solution

Since both parties dislike risk equally—both have the utility function $\ln(\cdot)$ —the principal will charge a very high price of insurance, so high that profit, π or p , will never be close to 0. In other words, the price of the coverage will—because of a risk-averse principal—be so high that even a risk-averse agent will find it ridiculous. The agent, even though being risk-averse, will therefore not agree to the contract

1 Moral Hazard (continued from PS8)

Consider the following principal-agent model:

There is a risk-neutral principal with utility function $U^P = x - w$, where $x \in \{2, 7\}$ is the principal's revenue and w the wage she pays the agent. The principal makes a take-it-or-leave-it-offer to the agent. If the agent accepts the offer, he chooses an effort level $e \in \{e^l, e^h\}$. His utility is $U^A = \sqrt{w} - c(e)$, where $c(e) = \sqrt{e}$. Note that $e^h > e^l \geq 0$. The effort level stochastically determines the principal's revenue:

$$Pr(x = 7|e^h) = \frac{3}{4} \text{ and } Pr(x = 7|e^l) = \frac{1}{2}.$$

If the agent rejects the offer, he receives his reservation utility $\underline{u} = 0$.

Suppose the agent's effort is unobservable and non-verifiable, i.e. solve the problem under the second best.

1. State the principal's problem, and explain each element (write at most 2 sentences per element).
2. Discuss the constraint(s) and whether it will be binding/non-binding.
3. Suppose that $e^h = 1$ and $e^l = \frac{4}{9}$. Derive first order conditions and discuss whether Lagrange multiplier(s) > 0 .
4. Suppose that $e^h = 1$ and $e^l = \frac{4}{9}$. Determine the optimal wage scheme for each possible effort level the principal may implement.
5. Suppose that $e^h = 1$ and $e^l = \frac{4}{9}$. Determine the optimal effort level.

1

Define $u(w_H) \equiv \sqrt{w_H}$ and $u(w_L) \equiv \sqrt{w_L}$

The principal's problem is to maximise the profit under two constraints: The PC and the IC:

The PC states that expected utility from taking the job must be greater than the utility the agent could get elsewhere, i.e., if P demands high effort

$$p_h(e^h)u(w_H) + (1 - p_h(e^h))u(w_H) - c(e^h) \geq \bar{u}$$

The IC states that the utility of performing a high effort must be greater than the increase in cost of performing this level of effort:

$$\begin{aligned} p_h(e^h)u(w_H) + (1 - p_h(e^h))u(w_L) - c(e^h) \\ \geq p_h(e^l)u(w_H) + (1 - p_h(e^l))u(w_L) - c(e^l) \end{aligned}$$

Use $1 - p_h(e^h) = p_l(e^h)$, $1 - p_h(e^l) = p_l(e^l)$ to rewrite the IC

$$(p_h(e^h) - p_h(e^l))u(w_H) + (p_l(e^h) - p_l(e^l))u(w_L) - (c(e^h) - c(e^l))$$

And the profit maximising is

$$\min p_h(e^h)w_H + (1 - p_h(e^h))w_L$$

3

The FOCs are found by first deriving the Lagrange

$$\begin{aligned}\mathcal{L} = & p_h(e^h)w_H + p_l(e^h)w_L \\ & + \lambda[p_h(e^h)u(w_H) + p_l(e^h)u(w_L) - c(e^h) \\ & - \bar{u}] + \mu \left[(p_h(e^h) - p_h(e^l))u(w_H) + (p_l(e^h) - p_l(e^l))u(w_L) \right. \\ & \left. - (c(e^h) - c(e^l)) \right]\end{aligned}$$

Now, taking the derivatives

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial w_L} &= p_h(e^h) + \lambda p_h(e^h)u'(w_H) + \mu(p_h(e^h) - p_h(e^l))u'(w_H) = 0 \\ \frac{\partial \mathcal{L}}{\partial w_H} &= p_l(e^h) + \lambda p_l(e^h)u'(w_L) + \mu(p_l(e^h) - p_l(e^l))u'(w_L) = 0 \\ \frac{\partial \mathcal{L}}{\partial \lambda} &= p_h(e^h)u(w_H) + p_l(e^h)u(w_L) - c(e^h) - \bar{u} = 0 \\ \frac{\partial \mathcal{L}}{\partial \mu} &= (p_h(e^h) - p_h(e^l))u(w_H) + (p_l(e^h) - p_l(e^l))u(w_L) - (c(e^h) - c(e^l)) = 0\end{aligned}$$

4

The two latter FOCs are two equations with two unknowns:

$$\begin{aligned}\frac{3}{4}w_H^{\frac{1}{2}} + \frac{1}{4}w_L^{\frac{1}{2}} - 1 &= 0 \\ \left(\frac{3}{4} - \frac{1}{2}\right)w_H^{\frac{1}{2}} + \left(\frac{1}{4} - \frac{1}{2}\right)w_L^{\frac{1}{2}} - \left(1 + \sqrt{\frac{4}{9}}\right) &= 0\end{aligned}$$

Which can be solved as

$$\begin{aligned}w_H &= \frac{16}{9} \\ w_L &= 0\end{aligned}$$

And if P wants to implement e_L , the solution is the same as in FB (page 42 in the book) so

$$\text{from PC: } w_L = w_H = e_L = \frac{4}{9}$$

5

$$\Pi_H = p_h(e^h)(\pi_H - w_H) + p_l(e^h)(\pi_L - w_L) = \frac{3}{4}\left(7 - \frac{16}{9}\right) + \frac{1}{4}(2 - 0) = 4.4167$$

$$\Pi_L = p_h(e^l)(\pi_H - w_H) + p_l(e^l)(\pi_L - w_L) = \frac{1}{2}\left(7 - \frac{4}{9}\right) + \frac{1}{2}\left(2 - \frac{4}{9}\right) = 4.0555$$

Implying that the optimal effort level is the high one because the profit is greater:

$$\Pi_H > \Pi_L$$

6. Suppose that $e^h = 3$ and $e^l = 1$. Can it be optimal for the principal to implement e^h ?
7. Suppose that $e^h = 1$ and $e^l = \frac{4}{9}$, but let $x \in \{2, 4, 7\}$. The probabilities that one of the three revenue levels realizes depends on the agent's effort as follows:

$$Pr(x = 4|e^h) = \frac{1}{2} \text{ and } Pr(x = 7|e^h) = \frac{1}{2}.$$

$$Pr(x = 2|e^l) = \frac{7}{9} \text{ and } Pr(x = 4|e^l) = \frac{1}{9} \text{ and } Pr(x = 7|e^l) = \frac{1}{9}.$$

Suppose the principal wants to implement e^h . What could he do to ensure that the agent chooses e^h ? (Hint: consider whether a fine paid out when $x=2$ could secure that the agent never chooses e^l .)

6

Since

$$\frac{3}{4} \cdot 7 + \frac{1}{4} \cdot 2 - e_H \geq \frac{1}{2} \cdot 7 + \frac{1}{2} \cdot 2 - e_L$$

I insert $e_H = 3, e_L = 1$

$$\frac{3}{4} \cdot 7 + \frac{1}{4} \cdot 2 - 3 \geq \frac{1}{2} \cdot 7 + \frac{1}{2} \cdot 2 - 1$$

Which is

$$2.75 \geq 3.5$$

Not satisfied

Something similar can be done with the IC and the same conclusion would be reached: the inequality is violated.

7

The PC:

$$w^{\frac{1}{2}} - e_H^{\frac{1}{2}} = \bar{u} = 0 \Leftrightarrow w_H = e_H$$

So,

$$u(w) - c(e^h) \geq u(w - f) - c(e^l)$$

That is,

$$\sqrt{w_H} - \sqrt{1} \geq \sqrt{w_H - f} - \sqrt{\frac{4}{9}} \Leftrightarrow \sqrt{1} - \sqrt{1} \geq \sqrt{1 - f} - \sqrt{\frac{4}{9}} \Leftrightarrow \frac{5}{9} \leq f \leq 1$$

Why is there an upper bound on the fine? Since we assume that no payoff can be negative

Problem 2: Insurance

Consider a risk-averse agent who owns an asset worth 100 USD. The agent can choose one of two actions: being cautious (a_1) or being negligent (a_0). The agent's payoff from having wealth w and taking action a is given by $u(w, a_0) = \sqrt{w}$ and $u(w, a_1) = \sqrt{w} - 3$. There are two possible states of nature: either damage occurs, in which case the asset is lost, or no damage occurs, in which case nothing happens to the asset. The probability that damage occurs equals 50% if the agent is negligent and 25% if he is cautious.

1. What is the optimal action for the agent?
2. Suppose there is a monopolistic firm that can make a take-it-or-leave-it offer (insurance) to the agent. Assume first that the agent's action is contractible. A contract in this case consists of a flat payment p the agent makes at the time of signing the contract, and the payout $d(\omega, a)$ in state ω after choosing action a . Thus, the agent's final wealth is the value of the asset depending on the state of nature plus the value of the contract $d(\omega, a) - p$. Write down the firm's maximization problem given that the agent chooses the action from question (1.). Check whether the constraint(s) are binding. What contract will the firm offer in this setting?

Hint: $d(w, a) = d(w)$. Argue why $d(w)$ is 0 when no damage occurs and $d(w) = d$ when damage occurs. Solve for d .

3. Suppose next that the agent's action is not contractible. In this case, the firm can only offer a contract consisting of a premium p and an reimbursement amount $d(\omega)$ that depends on the state ω . Write down the firm's maximization problem given that the firm wants the agent to choose the action from question (1.). What are the constraints on the firm's maximization problem? Check whether the constraint(s) are binding. What contract will the firm offer?

Hint: Formulate the appropriate IC constraint. Consider whether this constraint is likely to change the previous solution of the problem.

1

In the a_1 cautious scenario

$$u(w, a_1) = 0.75(\sqrt{100}) + 0.25(\sqrt{0}) - 3 = 4.5$$

And a_0 negligent

$$u(w, a_0) = 0.5(\sqrt{100}) + 0.5(\sqrt{0}) = 5$$

Agent should choose a_0

2

$$\max 0.5 \cdot p + 0.5 \cdot (p - d)$$

st (PC binding in optimum)

$$0.5(\sqrt{100 - p}) + 0.5(\sqrt{0 - p + d}) = 5 \Leftrightarrow p^* = d - \frac{1}{400}d^2$$

Insert the optimal p in the maximisation problem

$$\max 0.5 \cdot p + 0.5 \cdot (p - d) = \max 0.5 \cdot d - \frac{1}{400}d^2 + 0.5 \cdot \left(d - \frac{1}{400}d^2 - d\right)$$

FOC is

$$\begin{aligned} \frac{\partial \pi}{\partial d} &= 0.5 - \frac{1}{400} \cdot 2d + 0.5 \left(1 - \frac{1}{400}2d - 1\right) = 0.5 - \frac{1}{400} \cdot 2d + 0.5 \left(\frac{1}{400}2d\right) = 0 \Leftrightarrow d^* \\ &= 100 \end{aligned}$$

Price is then

$$p^* = d - \frac{1}{400}d^2 = 100 - \frac{1}{400} \cdot 100^2 = 75$$

And the principal thus offers full coverage which is efficient

3

It is efficient since the agent get highest possible coverage and the principal is optimising.

4

We get

$$\max 0.5 \cdot p + 0.5 \cdot (p - d)$$

St (PC and IC)

$$0.5(\sqrt{100 - p}) + 0.5(\sqrt{0 - p + d}) \geq 5$$

$$0.5(\sqrt{100 - p}) + 0.5(\sqrt{0 - p + d}) \geq 0.75(\sqrt{100 - p}) + 0.25(\sqrt{d - p}) - 3$$

The IC is probably not binding since it is just too costly for the agent to be negligent if the constraint is binding.

As an example, please note that the solution found in part 2 is not binding in IC since

$$0.5(\sqrt{100 - 75}) + 0.5(\sqrt{0 - 75 + 100}) \geq 0.75(\sqrt{100 - 75}) + 0.25(\sqrt{100 - 75}) - 3$$

Calculating LHS and RHS:

$$5 \geq 2$$

So not binding

3 Insurance

A rugby player (the agent) has an initial wealth of 100 DKK. The agent will experience a tooth injury resulting in a loss of l DKK. The amount of the loss varies and can take n values: $l \in \{l_1, \dots, l_n\}$, $l_i > 0$, $l_i < 100$ $i \in \{1, \dots, n\}$. The rugby player can either play rugby with low risk which requires a lot of concentration and attention ($a = a_h$) or he can play with high risk and low concentration ($a = a_l$). Attention $a = a_h$ costs the agent $c(a_h)$ and attention a_l costs him $c(a_l)$, $c(a_h) > c(a_l)$.

The severity of the injury is stochastically determined by the agent's attention:

$$Pr(l = l_i | a_h) = p_i^h \text{ and } Pr(l = l_i | a_l) = p_i^l,$$

where $\sum_{i=1}^n p_i^h l_i < \sum_{i=1}^n p_i^l l_i$, $\sum_{i=1}^n p_i^h = 1$, $\sum_{i=1}^n p_i^l = 1$ and $p_i^h > 0$ and $p_i^l > 0$ for all $i \in \{1, \dots, n\}$.

There is one risk-neutral insurance company (the principal), which makes a take-it-or-leave-it offer to the agent. If the agent pays π DKK to the principal, the principal guarantees to pay the agent $x(l_i)$ DKK (called the coverage) if a loss of l_i DKK occurs.

The utility function of the principal is $U^P = \pi - x(l_i)$. The utility function of the agent is additively separable in the utility from final wealth $\hat{w}$ and attention costs, i.e., $U^A = u(\hat{w}) - c(a)$. Furthermore we assume that the agent is risk averse in $\hat{w}$ such that $u''(\cdot) < 0$ and $u'(\cdot) > 0$.

Note: the final wealth for a realized loss l_i is $\hat{w} = 100 - l_i + x(l_i) - \pi$ if the agent buys insurance, and $\hat{w} = 100 - l_i$ if he does not buy insurance. Assume that if A does not buy insurance he always play rugby with a lot of attention and concentration ($a = a_h$).

1. Identify and discuss the potential "conflict of interest" in this problem. Why might the second best solution differ from the first best?
2. Write down the expected utility of the agent if he accepts the take-it- or-leave-it offer of the principal and provides attention level a_h .
3. (First best) Suppose the agent's attention level is observable and verifiable.
 - (a) State the principal's problem. Explain the different components of the problem.
 - (b) Explain how the principal determines whether he wants the contract to require attention level a_l or a_h ? And explain how this can be enforced in the contract?

2

Expected utility is

$$\sum_{i=1}^n p_i^h u(w - l_i + x(l_i) - \pi) - c(e^h)$$

3a

$$\max \sum_{i=1}^n p_i^j (\pi - x(l_i))$$

St

$$\sum_{i=1}^n p_i^j u(w - l_i + x(l_i) - \pi) - c(e^h) \geq \sum_{i=1}^n p_i^j u(w - l_i) - c(e^h) = \bar{u}$$

$j \in \{L, H\}$ which is determined by whether P wants high or low effort

3b

Depends on the profits associated with either level of attention. Sometimes, it is too costly to implement the high attention, and therefore, low attention is to be chosen.

3c and 3d

Principal's problem is

$$\max \sum_{i=1}^n p_i^h (\pi - x(l_i))$$

St

$$\sum_{i=1}^n p_i^h u(w - l_i + x(l_i) - \pi) - c(e^h) \geq \sum_{i=1}^n p_i^h u(w - l_i) - c(e^h) = \bar{u}$$

Lagrange:

$$\mathcal{L} = \sum_{i=1}^n p_i^h (\pi - x(l_i)) + \lambda \left(\sum_{i=1}^n p_i^h u(w - l_i + x(l_i) - \pi) - \sum_{i=1}^n p_i^h u(w - l_i) \right)$$

FOC:

$$\frac{\partial \mathcal{L}}{\partial \pi} = 1 + \lambda \sum_{i=1}^n p_i^h u'(w - l_i + x(l_i) - \pi) = 0$$

$$\frac{\partial \mathcal{L}}{\partial x(l_i)} = -p_i^h + \lambda(p_i^h u'(w - l_i + x(l_i) - \pi)) = 0 \Leftrightarrow \frac{1}{\lambda} = u'(w - l_i + x(l_i) - \pi)$$

Implying that marginal utility is the same in every scenario, hence full coverage: $x(l_i) = l_i$

- (c) Suppose the principal wants to implement a_h . Derive the first order conditions and interpret them. What does optimal behavior from the principal imply?
- (d) Suppose the principal wants to implement a_h . Specify the coverage and the price the principal specifies.
- 4. (Second best) Suppose from now on that the agent's attention level is unobservable and non-verifiable, and suppose the principal wants to implement a_h .
 - (a) Write down the Lagrangian and the first-order conditions.
 - i. State the principal's problem. Explain the different components of the problem.
 - ii. Use the first order conditions to find an equation that characterizes the optimal coverage $x(l_i)$ for every $i \in \{1, \dots, n\}$.
 - iii. Does the principal fully insure the agent? Why/why not? (2-3 sentences) Discuss this in relation to your answer under the First best.
 - (b) Give three examples of other real life situations where "informational asymmetries" and "conflict of interest" may lead to similar issues as those discussed above.

4a

i

Now, second best, so we must include an IC constraint in the optimisation problem

$$\max \sum_{i=1}^n p_i^h (\pi - x(l_i))$$

St (PC and IC)

$$\sum_{i=1}^n p_i^h u(w - l_i + x(l_i) - \pi) - c(e^h) \geq \sum_{i=1}^n p_i^h u(w - l_i) - c(e^h) = \bar{u}$$

$$\sum_{i=1}^n p_i^h u(w - l_i + x(l_i) - \pi) - c(e^h) \geq \sum_{i=1}^n p_i^l u(w - l_i + x(l_i) - \pi) - c(e^l)$$

IC can be simplified:

$$\sum_{i=1}^n (p_i^h - p_i^l) u(w - l_i + x(l_i) - \pi) - (c(e^h) - c(e^l))$$

Lagrange:

$$\mathcal{L} = \sum_{i=1}^n p_i^h (\pi - x(l_i)) + \lambda \left(\sum_{i=1}^n p_i^h u(w - l_i + x(l_i) - \pi) - \sum_{i=1}^n p_i^h u(w - l_i) \right) + \mu \left[\sum_{i=1}^n (p_i^h - p_i^l) u(w - l_i + x(l_i) - \pi) - (c(e^h) - c(e^l)) \right]$$

ii and iii

FOC

$$\frac{\partial \mathcal{L}}{\partial x(l_i)} = -p_i^h + \lambda (p_i^h u'(w - l_i + x(l_i) - \pi) + \mu (p_i^h - p_i^l) u'(w - l_i + x(l_i) - \pi)) = 0$$

$$\Leftrightarrow \frac{1}{u'(w - l_i + x(l_i) - \pi)} = \lambda + \mu \left(1 - \frac{p_i^l}{p_i^h} \right)$$

This implies that optimal coverage is dependent on the likelihood ratio (LR): $\frac{p_i^l}{p_i^h}$ small implies

that it is very likely that the agent paid high attention. Conversely, if $\frac{p_i^l}{p_i^h}$ is large, it is very

likely that the agent did not pay a lot of attention. So, because of this unobservable attention, the principal does not fully insure the agent

Why? A small LR implies that the probability of large severity of a loss given low attention is very large compared to the probability of a large severity of loss when high attention is provided if you interpret it as though a large p_i^j is a small possibility of a severe loss